SHOGUN-SYNC

Total War: Shogun 2 Multiplayer Synchronization Tool

Error Codes Reference

System Diagnostics and Troubleshooting Guide

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Module: Diagnostics (`errors.pdf`)

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1 Overview

Shogun-Sync uses a unified numbering format for all runtime errors:

```
Error <ID>:  <Description>
```

Example Terminal Output

```
Error 001: Save file not found
Error 015: Connection failed
Error 020: Password required
```

2 Error Codes Reference

Below is the complete reference table of all standardized error codes in Shogun-Sync.

Code	Message	File	Cause / Meaning
001	Save file not found	`kw_conf.c`, `kw_conn.c`	Target save file does not exist on disk.
002	File write failed	`kw_compress.c`	Cannot write file to disk (permissions / path).
003	Compression failed	`kw_compress.c`, `kw_conn.c`	Zlib deflate error during compression.
004	Compression init failed	`kw_compress.c`	Failed to initialize zlib deflate stream.
005	AppData path not found	`kw_conf.c`, `main.c`	Missing `%APPDATA%` variable in environment.
006	LocalAppData path not found	`kw_conf.c`	Missing `%LOCALAPPDATA%` variable.
007	Initialization failed	`kw_conf.c`, `main.c`	Could not create internal `.sync` folders.
008	Directory scan failed	`kw_conf.c`, `main.c`	Failed to read save files from directory.
009	Control handler failed	`main.c`	Could not register CTRL+C exit handler.
010	Memory allocation failed	`kw_compress.c`, `kw_conn.c`	System out of memory (`malloc` returned NULL).
011	Decompression init failed	`kw_compress.c`	Failed to initialize zlib inflate stream.
012	Decompression failed	`kw_compress.c`	Corrupted compressed data in inflate stream.
013	Decompression incomplete	`kw_compress.c`	Stream truncated before end of file.
014	Winsock init failed	`kw_conf.c`	Windows Sockets initialization error.
015	Connection failed	`kw_conn.c`	Cannot connect to lobby server via WebSocket.
016	Invalid packet received	`kw_conn.c`	Binary frame header is shorter than 64 bytes.

Code	Message	File	Cause / Meaning
017	Commit creation failed	`kw_conf.c`, `main.c`	Failed to create internal state commit.
018	Hostname resolution failed	`kw_conf.c`	Could not resolve local machine hostname.
019	Resync restore failed	`kw_conn.c`	Failed to decompress save received from host.
020	Command arguments error	`main.c`	Missing required room code or password in CLI.

3 Subsystem Explanations

3.1 Filesystem & Saves (001, 002, 005, 006, 007, 008)

These errors occur when the application interacts with the local filesystem:

- **Error 001:** Occurs when Shogun-Sync cannot find `autosave.save_multiplayer`. This usually means the campaign has not started yet or the game has not generated an autosave.
- **Error 005 & 006:** Standard Windows user environment variables (`APPDATA` / `LOCALAPPDATA`) could not be retrieved from the OS.
- **Error 007:** Triggered when running `shogun-sync init` if Windows blocks creating the hidden `.sync` directories.
- **Error 008:** Triggered if the save directory cannot be opened or enumerated.

3.2 Compression Engine (003, 004, 010, 011, 012, 013)

Handles compressing saves into blobs and decompressing them during auto-resync:

- **Error 003 & 004:** Compression errors during zlib deflate operations.
- **Error 011, 012 & 013:** Decompression errors. If the file transfer is interrupted over the network, zlib will reject the packet and protect the game folder from corrupt files.
- **Error 010:** The system does not have enough free RAM to hold the save buffer.

3.3 Network & Lobby (014, 015, 016, 018, 019, 020)

Handles multiplayer room creation, joining, and packet exchange:

- **Error 015:** Occurs when the program cannot reach the server (`wss://shogun2-sync.duckdns.org`) due to internet issues, an invalid room code, or a wrong password.
- **Error 016:** Received a malformed binary frame that does not contain the required 64-byte filename header.
- **Error 019:** The client received a binary save payload from the host, but decompression failed.
- **Error 020:** The user ran `create` or `join` without providing the necessary command line parameters.

4 Auto-Resync Workflow

When a desynchronization occurs during a multiplayer campaign, Shogun-Sync executes the following automated steps:

1. **Turn Detection:** Host generates a turn commit and broadcasts its SHA-1 hash to the lobby room (`<hash>|<tour>`).
2. **Hash Comparison:** The client compares its own autosave commit hash with the host's hash.
3. **Desync Trigger:** If the hashes differ, the client prints `Saves are desync` and automatically requests the host's save file via `REQ_SAVE|autosave.save_multiplayer`.

4. **Binary Transmission:** Host reads the save file, compresses it into memory, adds a 64-byte filename header, and transmits it via WebSocket binary frame.
5. **Local Restoration:** Client receives the binary payload, inflates it using `kw_decompress`, and replaces the desynchronized save on disk.
6. **Resolution:** Both players can now reload the save in Shogun 2 and continue their campaign.